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Editors-in-Chief
Energy and Buildings

Dear Editors,

I am pleased to submit the manuscript entitled *"EmuBAS: an open-source emulator of real-world building automation systems for testing building operation and artificial intelligence"* for consideration as a research article in *Energy and Buildings*.

New control sequences, fault detection and diagnosis, analytics and, increasingly, artificial intelligence agents, as well as the people who operate buildings, must ultimately perform in real buildings, where sensors drift, readings go missing, occupants behave randomly and equipment fails. Testing them in real buildings is slow, risky and without ground truth, while simulation test beds are usually idealised. The manuscript presents EmuBAS, an open-source emulator that couples a stepwise EnergyPlus simulation (the physics) with an explicitly modelled and fully labelled real-world layer in which the building automation system acts on imperfect observations, so that the physical consequences of imperfection are simulated. Faults and events from a library of 44 mathematically defined types can be injected at run time by people, scripts and AI agents; the clock runs from real time to 92,000 times real time; any EnergyPlus model is converted automatically; and the point list extends a typical building automation system with next-generation indoor-environment, weather and grid points. Using the DOE reference large office, the manuscript verifies the emulator against native EnergyPlus and demonstrates equipment-level fault impacts, labelled data for fault detection, AI agents in the loop, indoor air quality under wildfire smoke and the conversion of five reference buildings.

To my knowledge no comparable open tool exists, and its value reaches beyond algorithm development. Aviation and nuclear power train their operators on simulators, and autonomous-driving AI is developed in open simulators before it reaches the road; building operation has had no realistic equivalent. Because EmuBAS is safe to break, labelled, repeatable, fast and open, the manuscript identifies eighteen uses, from testing control sequences and commercial analytics to university teaching, operator training, commissioning practice and shared benchmarks.

I believe the work fits the scope of *Energy and Buildings* on building energy systems, controls, automation and data-driven methods, and that the open software and tutorials (`https://github.com/TensorBuildLab/Virtual-BAS`) will be useful to the journal's readership.

The manuscript is original, has not been published previously and is not under consideration

elsewhere. The author declares no competing interests.

Sincerely,

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